

**THIRD SEMESTER (CBCSS-UG) DEGREE EXAMINATION, NOVEMBER 2020****Mathematics****MTS 3B 03—CALCULUS OF SINGLE VARIABLE – 2****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 3B 03—CALCULUS OF SINGLE VARIABLE – 2  
(Multiple Choice Questions for SDE Candidates)

1.  $\int \frac{\ln x}{x} dx = \text{_____}.$

(A)  $\frac{1}{x^2} + c.$

(B)  $\frac{(\ln x)^2}{2} + c.$

(C)  $\frac{\ln x}{2} + c.$

(D)  $\frac{1}{x} - \ln x + c.$

2.  $\frac{d}{dn}(a^x) = \text{_____}.$

(A)  $xa^{x-1}.$

(B)  $a^x.$

(C)  $a^x \log a.$

(D)  $\frac{a^x}{\log a}.$

3.  $\int_1^{\sqrt{2}} x^{2x^2} dx = \text{_____}.$

(A) 2.

(B)  $\frac{1}{\ln 2}.$

(C)  $\frac{\ln 2}{2}.$

(D)  $\frac{2}{\ln 2}.$

4. Range of  $\cos hx$  is \_\_\_\_\_.

(A)  $[-1, 1].$

(B)  $(-\infty, \infty).$

(C)  $(-\infty, 1].$

(D)  $(-\infty, 1).$

5.  $\coth^{-1} x = \text{_____}$ .

(A)  $\frac{1}{\tanh x}$ .

(B)  $\tanh^{-1}\left(\frac{1}{x}\right)$ .

(C)  $\coth^{-1}\left(\frac{1}{x}\right)$ .

(D)  $\frac{1}{\tan^{-1}\left(\frac{1}{x}\right)}$ .

6.  $\lim_{x \rightarrow 1} \frac{1-x}{\ln x} = \text{_____}$ .

(A)  $-1$ .

(B)  $+1$ .

(C)  $0$ .

(D)  $2$ .

7.  $\lim_{x \rightarrow 0} \frac{a^x - b^x}{x} = \text{_____}$ .

(A)  $\ln\left(\frac{a}{b}\right)$ .

(B)  $\ln\left(\frac{b}{a}\right)$ .

(C)  $\ln(ab)$ .

(D)  $\infty$ .

8.  $\lim_{n \rightarrow \infty} \left(\frac{n-1}{n}\right) =$

(A)  $0$ .

(B)  $-1$ .

(C)  $1$ .

(D)  $\infty$ .

9.  $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = \text{_____}$ .

(A)  $0$ .

(B)  $1$ .

(C)  $\infty$ .

(D) None of these.

10. If  $|r| < 1$ ,  $\lim_{n \rightarrow \infty} r^n = \underline{\hspace{2cm}}$

(A) 0.

(B) 1.

(C)  $\infty$ .

(D) None of these

11. Let  $\{a_n\}$  be a sequence of positive terms such that  $a_n \geq a_{n+1}$  and  $\lim_{n \rightarrow \infty} a_n = 0$ , then :

(A)  $\sum a_n$  converges.

(B)  $\sum (-1)^n a_n$  converges.

(C)  $\sum a_n$  converges but  $\sum (-1)^n a_n$  diverges.

(D) None of these.

12. If the series  $\sum a_n (x - c)^n$  converges for  $x = k$ . Then it converges in :

(A)  $|r - c| < k$ .

(B)  $|r - c| \leq k$ .

(C)  $|r - c| > k$ .

(D)  $|r - c| \geq k$ .

13. The radius of convergence of the series  $\sum_{n=0}^{\infty} x^n$  is  $\underline{\hspace{2cm}}$ .

(A) 1.

(B) 2.

(C)  $\sqrt{2}$ .

(D)  $\infty$ .

14. Co-efficient of  $(x - 1)^2$  in the Taylor series expansion of  $f(x) = \frac{1}{x - 1}$  at  $x = 2$  is :

(A) -1.

(B)  $\frac{1}{2}$ .

(C)  $-\frac{1}{2}$ .

(D) +1.

15. Which of the following point lie on the curve  $r = -\sin\left(\frac{\theta}{3}\right)$ .

(A)  $\left(\frac{1}{2}, \frac{3\pi}{2}\right)$ .

(B)  $\left(\frac{1}{2}, \frac{\pi}{2}\right)$ .

(C)  $\left(1, \frac{3\pi}{2}\right)$ .

(D) None

16. Polar equation of a circle passing through origin, radius 2 and centre on +ve  $x$ -axis is :

(A)  $r = 4 \sin \theta$ .

(B)  $r = 4 \cos \theta$ .

(C)  $r = 2 \cos \theta$ .

(D)  $r = -2 \cos \theta$ .

17.  $\int_0^{\pi/6} \tan 2x \, dx = \underline{\hspace{2cm}}$ .

(A)  $\frac{1}{\ln 2}$ .

(B)  $\frac{1}{2} \ln 2$ .

(C)  $2 \ln 2$ .

(D)  $\frac{2}{\ln 2}$ .

18.  $\log_a(a^x) = \underline{\hspace{2cm}}$ .

(A)  $a^x$ .

(B)  $a$ .

(C)  $x^a$ .

(D)  $x$ .

19. The series  $\sum n^m x^n$  is converged if \_\_\_\_\_.

- (A)  $x > 1$  and  $x = 1$  when  $m < -1$ .      (B)  $x > 1$  and  $x = 1$  when  $m > -1$ .  
(C)  $x < 1$  and  $x = 1$  when  $m < -1$ .      (D)  $x < 1$  and  $x = 1$  when  $m > -1$ .

20.  $x = a \sec t, y = b \tan t$  is the parametric representation of :

- (A) Parabola.      (B) Ellipse.  
(C) Hyperbola.      (D) Circle.

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**THIRD SEMESTER (CBCSS-UG) DEGREE EXAMINATION, NOVEMBER 2020****Mathematics****MTS 3B 03—CALCULUS OF SINGLE VARIABLE – 2**

Time : Two Hours and a Half

Maximum : 80 Marks

**Section A***Answer at least ten questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 30.*

1. Find the derivative of  $y = e^{-x \cos x}$ .
2. Find the derivative of  $y = \log(e^{2x} + e^{-2x})$ .
3. Find  $\lim_{x \rightarrow 1^+} \frac{\sin \pi x}{\sqrt{x-1}}$ .
4. Show that  $\sinh^{-1} x = \log(x + \sqrt{x^2 + 1})$ .
5. Find  $\lim_{n \rightarrow \infty} e^{\sin(\frac{1}{n})}$ .
6. Express  $3.\overline{214}$  as a rational number.
7. Determine whether the series  $\sum_{n=1}^{\infty} \frac{1}{3+2^n}$  converges or diverges.
8. Show that the series  $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}+1}$  is divergent.
9. Find a power series representation of  $\frac{1}{(1-x)^2}$  on  $(-1, 1)$  by differentiating a power series representation of  $\frac{1}{1-x}$ .

10. Find the Maclaurin series of  $\sin x$  and determine its interval of convergence.
11. Find a rectangular equation whose graph contains the curve C with the given parametric equation  $x = 2t + 1, y = t - 3$ .
12. Find the equation of the tangent line to the curve  $x = \sec t, y = \tan t, -\frac{\pi}{2} < t < \frac{\pi}{2}$  at  $t = \frac{\pi}{4}$ .
13. Find the parametric equation for the line passing through the point  $(-2, 1, 3)$  and parallel to the vector  $\langle 1, 2, -2 \rangle$ .
14. The point  $\left(3, \frac{\pi}{3}, \frac{\pi}{4}\right)$  is expressed in spherical co-ordinates. Find its rectangular co-ordinates.
15. Find the antiderivative of  $r^1(t) = \cos t i + e^{-t} j + \sqrt{t} k$  satisfying the initial condition  $r(0) = i + 2j + 3k$ .  
(10 × 3 = 30 marks)

### Section B

*Answer at least five questions.*

*Each question carries 6 marks.*

*All questions can be attended.*

*Overall Ceiling 30.*

16. Use logarithmic differentiation to find the derivative of  $y = (\sqrt{\cos x})^x$ .
17. Evaluate  $\lim_{x \rightarrow 0} \frac{x^3}{x - \tan x}$ .
18. Evaluate  $\int_{-\infty}^0 x e^x dx$ .
19. Show that the series  $\sum_{n=1}^{\infty} \left[ \frac{2}{n(n+1)} - \frac{4}{3^n} \right]$  is convergent and find its sum.
20. Find the radius of convergence and interval of convergence of the series  $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2 \cdot z^n}$ .
21. Find the Taylor series for  $f(x) = \log x$  at  $x = 1$  and determine its interval of convergence.
22. Identify and sketch the surface  $4x - 3y^2 - 12z^2 = 0$ .
23. Let C be the helix  $r(t) = 2 \cos t i + 2 \sin t j + t k, t \geq 0$ . Find  $T(t)$  and  $N(t)$ .

(5 × 6 = 30 marks)

## Section C

Answer any **two** questions.  
Each question carries 10 marks.

24. (a) Find  $\int \frac{e^x}{e^{2x} + 1} dx$ .

(b) Find  $\int \cosh(2x + 3) dx$ .

(c) Evaluate  $\lim_{x \rightarrow 0} x^x$ .

25. (a) Evaluate  $\int_0^{\infty} \frac{e^{-\sqrt{x}}}{\sqrt{x}} dx$ .

(b) Determine whether the series  $\sum_{n=1}^{\infty} \frac{2n^2 + n}{\sqrt{4n^7 + 3}}$  converges or diverges.

26. Find the area of the region that lies outside the circle  $r = 3$  and inside the cardioid  $r = 2 + 2\cos\theta$ .

27. A shell is fired from a gun located on a hill 100 m above a level terrain. The muzzle speed of the gun is 500 m/sec and its angle of elevation is  $30^\circ$ .

(a) Find the range of the shell.

(b) What is the maximum height attained by the shell?

(c) What is the speed of the shell at impact?

(2 × 10 = 20 marks)